

BBO capstone project presentation template

Develop a concise presentation that captures:

1. An overview of your BBO approach

- Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
- Briefly describe your overall strategy and how your process fits together.

I used this BBO capstone to learn how to maximize unknown functions with a limited query budget. The core goal was to discover high-value regions efficiently instead of relying on brute-force search.

My overall process was: start from seed points, combine them with all previous submissions as a Gaussian Process surrogate model, and then refine promising regions. Each round became a better informed search rather than an isolated guess.

2. How your strategy has evolved

- Describe the key changes in your approach since your early rounds.
- What influenced these changes – data trends, feedback, model performance or intuition?
- Identify which principles or heuristics now guide your query decisions.

Early rounds were broad and exploratory. I sampled widely to detect signal, understand scale, and see which functions were sparse or highly sensitive. Later rounds became more systematic. Data trends, weak outputs, and model behavior pushed me toward significant regions, seeded GP fits, and reproducible query generation. I also connected notebooks that were implicitly minimizing when the project regeneration maximization. My current heuristics are simple: stay near the top-performing points, reduce unnecessary global jumps, and only explore when the uncertainty looks genuinely informative.

3. Patterns, data and insights

- Highlight the most meaningful trends you've observed across your data points or model evaluations.
- Identify which variables or behaviours seem to influence results the most.
- Explain how these observations have shaped your understanding of the search process.

Across the data, the clearest pattern is region quality. Results improve sharply when a query lands inside a productive area and drop quickly when the search drifts away. Functions 1, 5 and 8 showed strong local structure, while Functions 4, 6 and 7 showed how expensive it is to search in the wrong region.

The biggest drivers of variation were locality, boundary behavior, and how much the strongly the surrogate rewarded uncertainty. P-OA-like behavior, these behave these the way like the main components explaining most of the performance differences across rounds.

4. Decision-making and iteration

- Explain how you balance exploration and exploitation.
- Describe one or two examples of strategic decisions. What worked, what didn't, and why?
- Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

I balance exploration and exploitation by stage. Early rounds use exploration to find signal; later rounds favor exploitation around the best-known region. One decision that worked well was tightening the trust region for Functions 1, 5, and 8, which improved the best values. One that failed was allowing Functions 4, 5, and 6 to behave like randomization problems; that pulled queries toward poor regions until I fixed the objective log minimization problems; that pulled queries toward poor regions until I fixed the objective log.

When results do not match expectations, I compare the latest point with the best-so-far region, inspect the search behavior, and adapt the notebook rather than trusting the previous heuristic.

5. Next steps and reflection

- Outline your planned actions or experiments to improve the model's performance.
- Include two or three sentences connecting your work to the broader ML landscape.
- Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

For the final round, I plan to use function-specific trust regions, keep only limited exploration, and focus most queries around the top-performing points. I also want each query to test a clear hypothesis rather than adding random coverage.

This project connects strongly to the broader ML landscape because it uses surrogate models, model uncertainty, and iterative learning under limited data. The same ideas appear in hyperparameter tuning, experimental design, and expensive real-world optimization.

To a non-technical stakeholder, I would say: we are using each new result to narrow the search, avoid wasted trials, and move step by step toward the best-performing settings.